

Gamma controls the scales available to readout gradient descent

Research note Mechanism, measured delays, and self-contained proof

Gamma changes how quickly a readout learns an attainable target. For fixed neuron centers, a common slope applies an explicit frequency filter to the features and hence to the kernel governing readout updates. Small gamma strongly attenuates fine spatial variation. Carrying this filter through the finite training geometry predicts the observed delay: reducing gamma from 64 to 8 requires **986.59 times as many updates** for the primary target. Across five targets, measured slowdowns range from 3.34 to 986.59; all 19 available executed acquisition times lie inside the predicted intervals, with one additional run censored. The mechanism is exact; the delay depends on the target, geometry, and optimizer.

1 The kernel gives gamma a direct role in optimization

Freeze the common slope $\gamma > 0$ and centers c_j in the model

$$f_{\gamma, \theta}(x) = b + \sum_{j=1}^W w_j \tanh(\gamma(x - c_j)), \quad \theta = (b, w_1, \dots, w_W). \quad (1)$$

Train only the raw readout coefficients from zero on half empirical mean squared error at m training inputs x_i . Compare slopes with the same centers, inputs, target, coefficient coordinates, and specified step-size rule. For the feature vector $\phi_\gamma(x) = (1, \tanh(\gamma(x - c_1)), \dots, \tanh(\gamma(x - c_W)))^T$, define

$$k_\gamma(x, x') = \phi_\gamma(x)^T \phi_\gamma(x'), \quad (K_\gamma)_{i\ell} = k_\gamma(x_i, x_\ell) / m. \quad (2)$$

Writing r_n for target values minus predictions after n updates, ordinary readout gradient descent (GD) satisfies

$$\boxed{r_{n+1} = (I - \eta_\gamma K_\gamma) r_n.} \quad (3)$$

Thus the kernel describes which errors a coefficient update corrects together. Similar feature responses make two predictions move together; learning their difference can be slow. Gamma's optimization effect follows from how it changes this update matrix.

2 An explicit gamma filter

The fixed reference dictionary consists of sharp steps at the same centers:

$$k_{\text{step}}(x, x') = 1 + \sum_{j=1}^W \text{sign}(x - c_j) \text{sign}(x' - c_j), \quad (4)$$

$$\rho_\gamma(t) = \frac{\gamma}{2} \text{sech}^2(\gamma t).$$

The density ρ_γ averages a sharp step into a tanh transition with width proportional to $1/\gamma$. The following theorem identifies the resulting filter exactly.

Theorem 1 (Gamma-controlled smoothing). For any fixed centers and common slope $\gamma > 0$, the tanh readout kernel is the step kernel averaged over both inputs:

$$k_\gamma(x, x') = \iint_{\mathbb{R}^2} \rho_\gamma(x - t) \rho_\gamma(x' - s) k_{\text{step}}(t, s) dt ds. \quad (5)$$

In either input, averaging multiplies a frequency- ω component by

$$M_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_\gamma(0) = 1. \quad (6)$$

A common-slope cap $0 < \gamma \leq \Gamma$ implies $M_\gamma(\omega) \leq M_\Gamma(\omega)$. Frequencies well below gamma are nearly preserved; those well above gamma have exponentially small multipliers, $M_\gamma(\omega) \sim 2ze^{-z}$ as $z \rightarrow \infty$.

The mechanism is selective attenuation of spatial scales. Reducing gamma from 64 to 8 changes the multiplier at $\omega = 2\pi$ from 0.9960 to 0.7851, but at $\omega = 10\pi$ from 0.9074 to 0.02584. The finer component's multiplier becomes about 35.1 times smaller. Gamma changes the relative strength of corrections even when the step compensates for overall kernel magnitude. Appendix A gives the full proof, including the transform calculation. No periodic sample or center assumption is needed.

Finite sampling and center geometry determine how the filtered components combine into learning directions. Fourier waves need not be eigenvectors of the sampled kernel, so M_γ^2 is generally neither a finite-kernel eigenvalue nor an inverse learning time. We retain these couplings when computing the prediction. The cap guarantees feature attenuation; the target's interaction with the resulting kernel determines the delay.

A small example with perfect capacity

Take one centered neuron, samples $-s, +s$ with $s > 0$, and labels $-1, +1$. Every positive gamma fits exactly with $b = 0$ and $w = 1/\tanh(\gamma s)$. Yet at step $\eta = 0.5$, reaching 1% residual takes **925 updates** when $\gamma s = 0.1$ and **14 updates** when $\gamma s = 1$. The largest kernel eigenvalue is one in both cases. The target's contrast direction has eigenvalue $\tanh^2(\gamma s)$, explaining the different rates. Appendix B gives its exact acquisition count and the general GD corollary.

What the theorem contributes to the quantitative explanation

The theorem isolates gamma in a known frequency filter. Applying that filter to the fixed finite geometry and target gives a learning curve through standard least-squares GD. A controlled finite expansion makes the calculation practical and supplies acquisition intervals. No rate is fitted to an optimization trajectory. The following figures test the calculation retrospectively against archived training runs.

3 The filter retains the measured timing accuracy

The primary problem has $W = 559$ hidden neurons, $m = 8,193$ equally spaced training inputs on $[-1, 1]$, equispaced centers with a boundary halo, and target

$$f^*(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x). \quad (7)$$

Readouts start at zero. Each saved step is approximately $\eta_\gamma = 0.5/L_\gamma$, where L_γ is the largest eigenvalue of K_γ . All four models actually attain 1% relative residual, equivalent to relative squared loss 10^{-4} . Their different acquisition times therefore demonstrate an optimization effect at an attainable accuracy.

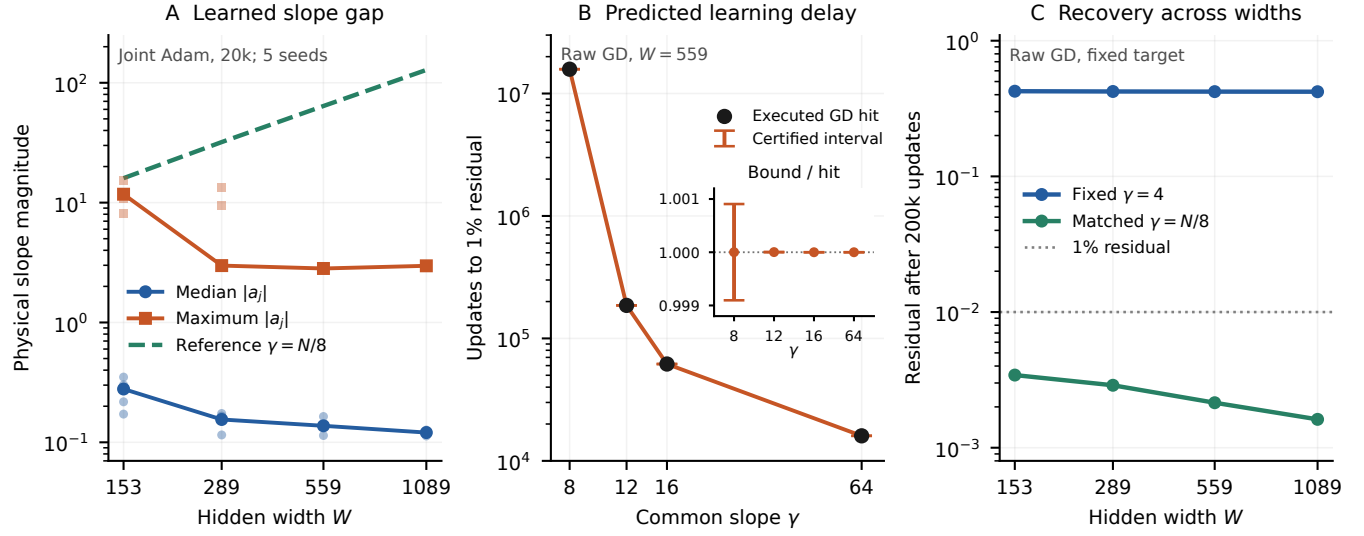


Figure 1: Observed scale gap, predicted delay, and a slope intervention. All panels use the sine-mixture target. **A:** After 20,000 joint Adam updates, faint points show the median and maximum absolute hidden slope for each of five seeds; solid curves show their medians across seeds. The dashed $\gamma = N/8$ construction reference is sufficient in that construction, not an asserted necessary threshold. **B:** Filter-derived intervals match executed zero-initialized raw readout GD hits at 1% residual for 559 fixed-center features. The inset divides each interval by its observed hit. These are deterministic bounds. **C:** After 200,000 raw readout updates, fixed gamma 4 leaves residual near 0.42, while gamma $N/8$ gives 0.0034–0.0016 across widths 153–1089. Panels B/C use curvature-normalized steps; A motivates the scale question from joint training. The theorem concerns common-slope frozen features.

The measured first-hit time changes from 15,798,313 updates at gamma 8 to 16,013 at gamma 64. The predicted delay ratio is **985.70–987.49**, enclosing the executed **986.59**. At gamma 8, the interval’s total width is 0.181% of the observed time. These primary endpoints have independent interval-arithmetic certificates.

A scalar control that rescales only the gamma-64 kernel to match each case’s largest eigenvalue predicts 16,013 updates at every gamma under the saved step rule. The full frequency-dependent filter accounts for the large delay this magnitude-only control misses. The numerical intervals are tight for the specified geometry and target; the theorem does not assert a universal slowdown factor or monotone learning time for every target.

4 The delay depends on the target

Four control targets use the same geometry and four slopes: the single sine $\sqrt{2} \sin(2\pi x)$, Runge function $1/(1 + 25x^2)$, quadratic $\sqrt{5}x^2$, and exponential of sine $\exp(\sin(3\pi x))$. Every target reaches 1% at gamma 8 and gamma 64, allowing the measured comparison below.

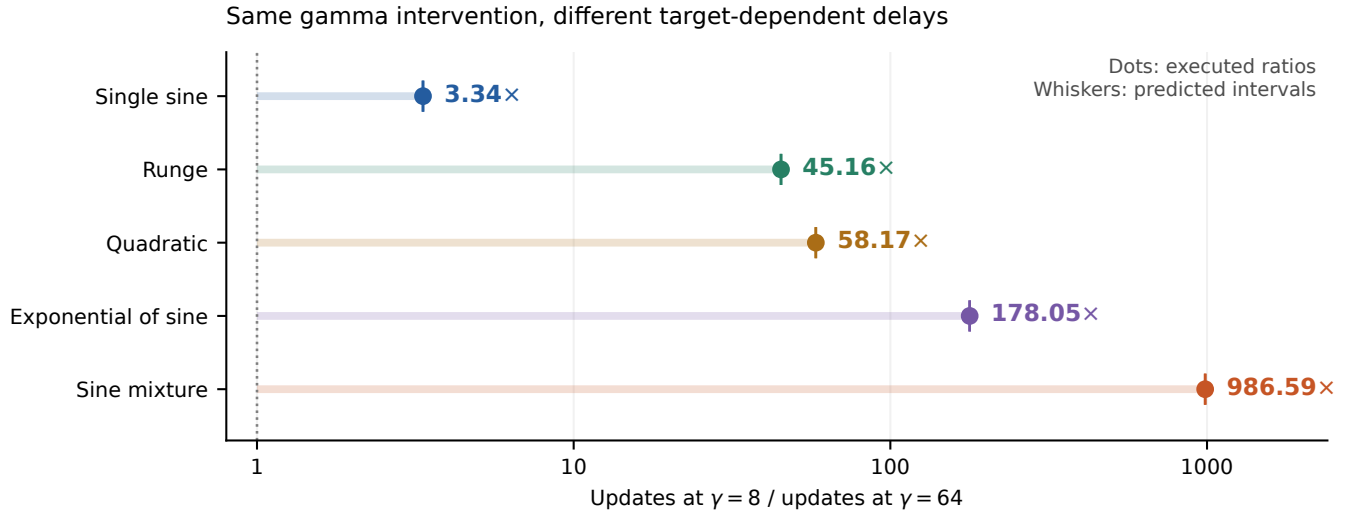


Figure 2: The same gamma intervention gives target-dependent delays. Each dot is the executed first-hit count at gamma 8 divided by that at gamma 64. Whiskers show the ratio interval from the predicted acquisition bounds and are narrower than the dots at this scale. Samples, centers, zero initialization, 1% residual tolerance, and curvature-normalized step rule are shared. Only the sine-mixture endpoints have independent interval-arithmetic certification; the controls have numerical sensitivity checks. Figure 3 includes every intermediate gamma and magnifies interval tightness.

The same filter produces very different delays because each target places different energy in the finite kernel’s learning directions. These observations support a quantitative target-dependent explanation. The quadratic still needs 58.17 times as many updates at gamma 8: low polynomial degree alone does not imply fast acquisition in this tanh dictionary. Finite geometry and target alignment remain part of the calculation.

What scaling gamma with width recovers

Figure 1C tests an intervention suggested by the filter: increase gamma as center spacing shrinks to preserve access to finer spatial scales. Gamma proportional to inverse spacing keeps the multiplier unchanged at frequencies measured relative to that spacing. In this construction, that means gamma proportional to width; Appendix C states this precisely.

Panel C holds the physical target fixed. It demonstrates a finite-budget benefit, and its residual near 0.42 is not an established irreducible floor. The filter identity does not by itself guarantee constant training times across widths. All figures concern training behavior, with no held-out generalization claim.

A Complete proof of the gamma mechanism

The density in (4) has unit mass and cumulative integral $\int_{-\infty}^a \rho_\gamma(t) dt = [1 + \tanh(\gamma a)]/2$. Splitting at a step's center gives

$$\int_{\mathbb{R}} \rho_\gamma(x-t) \operatorname{sign}(t-c) dt = \tanh(\gamma(x-c)). \quad (8)$$

Insert the finite sum defining k_{step} into the double integral in (5). Each pair of steps becomes a pair of tanh features, and the constant stays one. Bounded features and an integrable density justify all exchanges; the steps themselves need not be integrable.

Use Fourier convention $\widehat{\rho}(\omega) = \int_{\mathbb{R}} \rho(t) e^{-i\omega t} dt$. For a real number $a > 0$, define

$$I(a) = \int_{\mathbb{R}} \frac{e^{-iau}}{\cosh^2 u} du, \quad F(\zeta) = \frac{e^{-ia\zeta}}{\cosh^2 \zeta}. \quad (9)$$

Integrate F counterclockwise around the rectangle with vertices $-R, R, R+i\pi, -R+i\pi$, then let $R \rightarrow \infty$. On either vertical side, $|e^{-ia\zeta}| \leq e^{a\pi}$ and $|\cosh(\pm R + i\nu)|^2 = \sinh^2 R + \cos^2 \nu \geq \sinh^2 R$ for $0 \leq \nu \leq \pi$, so those integrals vanish. Since $F(u+i\pi) = e^{a\pi} F(u)$ and the upper side runs right to left, the horizontal integrals tend to $(1 - e^{a\pi})I(a)$.

The only enclosed pole is the double pole $\zeta_0 = i\pi/2$. Set $w = \zeta - \zeta_0$. Using $\cosh(\zeta_0 + w) = i \sinh w$,

$$F(\zeta_0 + w) = e^{a\pi/2} \left(-\frac{1}{w^2} + \frac{ia}{w} + O(1) \right), \quad \operatorname{Res}_{\zeta=\zeta_0} F = ia e^{a\pi/2}. \quad (10)$$

The residue theorem yields

$$(1 - e^{a\pi})I(a) = -2\pi a e^{a\pi/2}, \quad I(a) = \frac{\pi a}{\sinh(\pi a/2)}. \quad (11)$$

The sine part of the real-line integrand is odd, so I is real and even; the formula extends to negative a . At zero, $I(0) = \int_{\mathbb{R}} \operatorname{sech}^2 u du = 2$, the continuous limit. Substituting $u = \gamma t$ gives the normalized multiplier

$$\widehat{\rho}_\gamma(\omega) = \frac{1}{2} I(\omega/\gamma) = \frac{\pi|\omega|/(2\gamma)}{\sinh(\pi|\omega|/(2\gamma))} = M_\gamma(\omega), \quad M_\gamma(0) = 1. \quad (12)$$

In particular,

$$\int_{\mathbb{R}} \rho_\gamma(x-t) e^{i\omega t} dt = M_\gamma(\omega) e^{i\omega x}. \quad (13)$$

This proves the frequency interpretation in either kernel input without requiring an ordinary integrable Fourier transform of the steps. Expanding $\sinh z$ gives $M_\gamma = 1 - z^2/6 + O(z^4)$ at small z ; $\sinh z \sim e^z/2$ gives $M_\gamma \sim 2ze^{-z}$ at large z .

Finally, the derivative of $z/\sinh z$ has numerator $\sinh z - z \cosh z < 0$ for $z > 0$: this numerator vanishes at zero and has derivative $-z \sinh z < 0$. Reducing gamma increases z , proving the cap inequality. This completes Theorem 1.

B Standard GD corollary and acquisition intervals

Let $y_i = f^*(x_i)$ be the nonzero target vector, and let Φ_γ have rows $\phi_\gamma(x_i)^T$. Thus $K_\gamma = \Phi_\gamma \Phi_\gamma^T / m$. From zero initialization, with $0 < \eta_\gamma \leq 1/L_\gamma$, define

$$E_\gamma(n) = \frac{\|(I - \eta_\gamma K_\gamma)^n y\|}{\|y\|}, \quad n_\epsilon(\gamma) = \min\{n \in \mathbb{N}_0 : E_\gamma(n) \leq \epsilon\}. \quad (14)$$

Here $0 < \epsilon < 1$; an empty crossing set has time $+\infty$. This is a kernel-and-target calculation requiring no executed GD trajectory.

Corollary 1 (Target-dependent acquisition). Let $u_i(\gamma)$ be orthonormal eigenvectors for the positive eigenvalues $\lambda_i(\gamma)$ of K_γ , with $p_i(\gamma) = |u_i(\gamma)^T y|^2 / \|y\|^2$. Let $E_{\text{floor}}(\gamma)$ be the minimum relative residual over all readouts in this same tanh model. Then

$$E_\gamma(n)^2 = E_{\text{floor}}(\gamma)^2 + \sum_i p_i(\gamma) (1 - \eta_\gamma \lambda_i(\gamma))^{2n}. \quad (15)$$

Proof. Expand y in an orthonormal eigenbasis of the symmetric kernel. Nullspace components remain unchanged and give the least-squares floor. Every positive mode decays by the corresponding factor in (15). Orthogonality adds their squared residual norms. $\square$

The weights are target energy fractions, or squared cosine alignments with unit eigenvectors. Gamma changes the rates and these weights through the explicit filter. Eigenvalues λ_i are distinct from the dimensionless bandwidth β in Appendix C.

In the two-point example, the target occupies the contrast mode with eigenvalue $\tanh^2(\gamma s)$ and zero floor. For $0 < \eta < 1$,

$$n_\epsilon(\gamma) = \left\lceil \frac{\log \epsilon}{\log(1 - \eta \tanh^2(\gamma s))} \right\rceil \sim \frac{\log(1/\epsilon)}{\eta \gamma^2 s^2} \quad (\gamma s \rightarrow 0). \quad (16)$$

For the full experiment, a finite harmonic expansion retains the sample/center couplings and bounds omitted terms, including distant transitions in its auxiliary periodic representation. It approximates the original nonperiodic tanh model with the original readout coordinates. The detailed construction is in the [technical supplement](#).

If $\widehat{E}_\gamma(n)$ is the predicted residual and a justified margin satisfies $|E_\gamma(n) - \widehat{E}_\gamma(n)| \leq d_\gamma(n)$, then integer witnesses a, b give

$$\begin{aligned} \widehat{E}_\gamma(a) - d_\gamma(a) &> \epsilon, & \widehat{E}_\gamma(b) + d_\gamma(b) &\leq \epsilon \\ \implies a + 1 &\leq n_\epsilon(\gamma) \leq b. \end{aligned} \quad (17)$$

The true residual is nonincreasing under the step condition, proving the necessary count; the second witness proves sufficiency. The simplest margin is $n\eta_\gamma \Delta$ for kernel error bound Δ , with both update operators contracting. The refinement retains the error's action on the target. Dividing necessary and sufficient counts across gammas gives the ratio interval. Numerical verification status is separate from these exact-arithmetic statements (Appendix E).

C Width scaling preserves relative bandwidth

For center spacing h , set $\beta = \gamma h$ and grid-relative frequency $\xi = h\omega$. Fixing $\beta > 0$ gives

$$M_{\beta/h}(\xi/h) = \frac{\pi|\xi|/(2\beta)}{\sinh(\pi|\xi|/(2\beta))}, \quad (18)$$

independent of h , with value one at zero. For fixed gamma and nonzero ξ , as $h \rightarrow 0$,

$$M_{\gamma}(\xi/h)^2 \sim \frac{\pi^2 \xi^2}{\gamma^2 h^2} \exp\left(-\frac{\pi|\xi|}{\gamma h}\right). \quad (19)$$

The construction uses $h = 2/N$, $W = N + 2\lceil\sqrt{N}\rceil + 1$, and $\gamma = N/8$, hence $\beta = 1/4$ and $\gamma = \Theta(W)$. This avoids a growing exponential attenuation at the dictionary's resolution. An upper bound $\gamma = O(W)$ alone also permits fixed gamma and does not imply the conclusion. At fixed physical frequency, fixed gamma already gives width-independent attenuation; linear scaling makes its multiplier approach one. Geometry and target weights still determine training time.

D Protocol and evidence provenance

Figure 1A trains hidden slopes, offsets, and readout parameters for 20,000 Adam updates, with learning rate 10^{-3} , epsilon 10^{-8} , and moments (0.9, 0.999). Hidden slopes and readout weights start independently uniform on $[-\sqrt{6/(W+1)}, \sqrt{6/(W+1)}]$; biases start at zero. Seeds 0–4 are all shown. The width study uses $N = 128, 256, 512, 1024$; the acquisition study uses $N = 512$. Grids have $m = 16N + 1$ equally spaced endpoint samples on $[-1, 1]$. Frozen centers have spacing $2/N$ with a halo of $\lceil\sqrt{N}\rceil$ additional centers at each end.

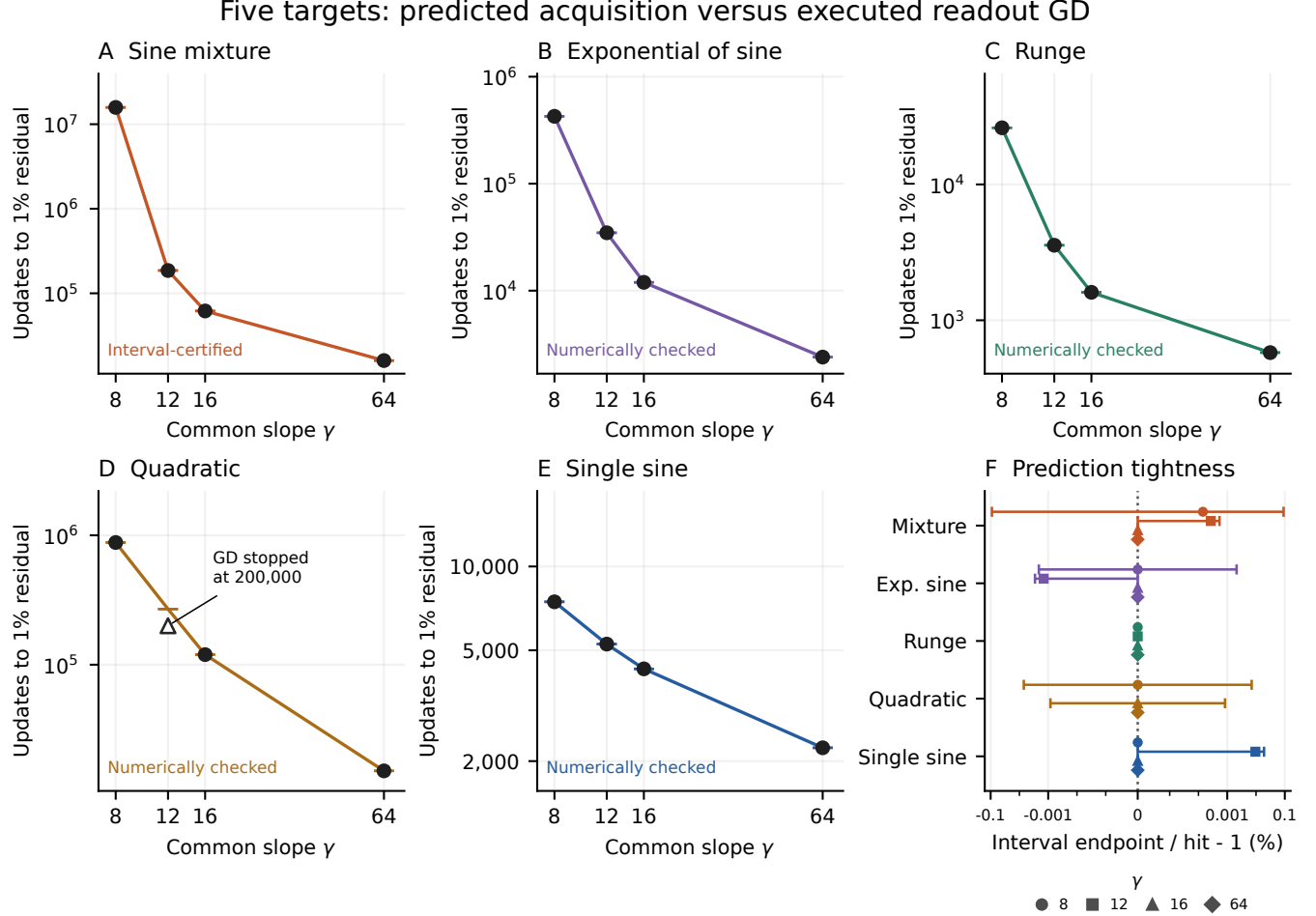
In Figure 1C, the target, inputs, centers, raw coordinates, and 200,000-update budget are shared within each width. Gamma is the intervention, and the step follows each kernel's curvature. These are measured residuals; certified times have not been computed at every width. Joint-training observations in A do not establish a theorem about joint Adam dynamics.

The four primary intervals have independent 192-bit interval-arithmetic endpoint certificates for nominal-real tanh on archived binary inputs and saved steps. They do not certify every floating-point iterate or plotted curve. The sixteen control intervals have double-precision reference and arithmetic-sensitivity checks, including a disclosed heuristic allowance for transcendental evaluations. No separate interval certificates were computed for those targets. A zero-width numerical interval alone does not certify an exact first hit.

Review materials. The [companion Markdown note](#) and [technical supplement](#) provide the narrative and full numerical construction. The [evaluation report](#) includes control targets, the scalar-rescaling control, analytic versus combined margins, and sensitivity checks. The [numerical summary](#), [interval audit](#), and [figure data](#) preserve exact values and verification status. Figures reuse archived results; no new training was needed. The [validation record](#) separates prior implementation tests from this document's checks.

E Complete five-target acquisition results

All cases share the 559-neuron geometry, 8,193 samples, zero raw readout, half empirical mean squared error, and each gamma's saved step, approximately $0.5/L_\gamma$. Intervals combine the analytic and target-action margins. Harmonic resolution is selected from calculated bounds, without fitting endpoints to executed hits.



A-E: colored lines and whiskers are predictions; black dots are executed hits. Axes use separate vertical ranges. F: all 19 executed hits; symmetric-log horizontal scale.

Figure 3: All twenty predictions, with nineteen executed crossings. A–E: Colored curves and whiskers give predicted first-hit intervals; black dots give executed hits at 1% residual. Vertical ranges differ; connecting lines guide the eye. The open triangle marks the quadratic run stopped at 200,000 updates and is not an observed crossing. **F:** Each interval endpoint is divided by its executed hit, shifted by one, and multiplied by 100. The symmetric-log axis expands small deviations. All 19 intervals contain zero; the censored case is excluded. The largest total relative width is 0.181%, for the sine mixture at gamma 8. Only A's intervals have independent interval-arithmetic certificates.

The censored quadratic run at gamma 12 has a predicted crossing interval 269,286–269,299 and a separate original-kernel spectral forecast of 269,292. Both remain forecasts. The other nineteen cases compare predictions against actual GD crossings. These checks assess the target-dependent timing calculation on archived training problems; they do not establish a universal learning-time ordering over targets or dictionaries.